Advanced Python Programming

23MID0233

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Exploring Visual Capitalist-using pivot_table

In [46]:
```
import pandas as pd

raw_data = [
    {"Rank": 1, "City": "Tampa", "Country": "USA", "GVI": 36.1},
    {"Rank": 2, "City": "Breda", "Country": "Netherlands", "GVI": 29.3},
    {"Rank": 3, "City": "Singapore", "Country": "Singapore", "GVI": 29.3}
    {"Rank": 4, "City": "Oslo", "Country": "Norway", "GVI": 28.8},
    {"Rank": 5, "City": "Sydney", "Country": "Australia", "GVI": 25.9},
    {"Rank": 6, "City": "Vancouver", "Country": "Canada", "GVI": 25.9},
    {"Rank": 7, "City": "Montreal", "Country": "Canada", "GVI": 25.5},
    {"Rank": 8, "City": "Cambridge", "Country": "USA", "GVI": 25.3},
    {"Rank": 9, "City": "Xalapa", "Country": "Mexico", "GVI": 24.0},
    {"Rank": 10, "City": "Durban", "Country": "South Africa", "GVI": 23.7
    {"Rank": 11, "City": "Johannesburg", "Country": "South Africa", "GVI"
    {"Rank": 12, "City": "Sacramento", "Country": "USA", "GVI": 23.6},
    {"Rank": 13, "City": "Frankfurt", "Country": "Germany", "GVI": 21.5},
    {"Rank": 14, "City": "Geneva", "Country": "Switzerland", "GVI": 21.4}
    {"Rank": 15, "City": "Veracruz", "Country": "Mexico", "GVI": 21.4},
    {"Rank": 16, "City": "Amsterdam", "Country": "Netherlands", "GVI": 20
    {"Rank": 17, "City": "Seattle", "Country": "USA", "GVI": 20.0},
    {"Rank": 18, "City": "Toronto", "Country": "Canada", "GVI": 19.5},
    {"Rank": 19, "City": "Miami", "Country": "USA", "GVI": 19.4},
    {"Rank": 20, "City": "Boston", "Country": "USA", "GVI": 18.2},
    {"Rank": 21, "City": "Tel Aviv", "Country": "Israel", "GVI": 17.5},
    {"Rank": 22, "City": "Guadalajara", "Country": "Mexico", "GVI": 17.3}
    {"Rank": 23, "City": "León", "Country": "Mexico", "GVI": 16.5},
    {"Rank": 24, "City": "Turin", "Country": "Italy", "GVI": 16.2},
    {"Rank": 25, "City": "Los Angeles", "Country": "USA", "GVI": 15.2},
    {"Rank": 26, "City": "Buenos Aires", "Country": "Argentina", "GVI": 1
    {"Rank": 27, "City": "Franca", "Country": "Brazil", "GVI": 13.7},
    {"Rank": 28, "City": "New York", "Country": "USA", "GVI": 13.5},
    {"Rank": 29, "City": "Cape Town", "Country": "South Africa", "GVI": 1
    {"Rank": 30, "City": "London", "Country": "United Kingdom", "GVI": 12
    {"Rank": 31, "City": "São Paulo", "Country": "Brazil", "GVI": 11.7},
    {"Rank": 32, "City": "Quito", "Country": "Ecuador", "GVI": 10.8},
    {"Rank": 33, "City": "Kobe", "Country": "Japan", "GVI": 9.4},
    {"Rank": 34, "City": "Paris", "Country": "France", "GVI": 8.8}
]

# Convert the array into DataFrame
df = pd.DataFrame(raw_data)
df
```

Out[46]:

	Rank	City	Country	GVI
0	1	Tampa	USA	36.1
1	2	Breda	Netherlands	29.3
2	3	Singapore	Singapore	29.3
3	4	Oslo	Norway	28.8
4	5	Sydney	Australia	25.9
5	6	Vancouver	Canada	25.9
6	7	Montreal	Canada	25.5
7	8	Cambridge	USA	25.3
8	9	Xalapa	Mexico	24.0
9	10	Durban	South Africa	23.7
10	11	Johannesburg	South Africa	23.6
11	12	Sacramento	USA	23.6
12	13	Frankfurt	Germany	21.5
13	14	Geneva	Switzerland	21.4
14	15	Veracruz	Mexico	21.4
15	16	Amsterdam	Netherlands	20.6
16	17	Seattle	USA	20.0
17	18	Toronto	Canada	19.5
18	19	Miami	USA	19.4
19	20	Boston	USA	18.2
20	21	Tel Aviv	Israel	17.5
21	22	Guadalajara	Mexico	17.3
22	23	León	Mexico	16.5
23	24	Turin	Italy	16.2
24	25	Los Angeles	USA	15.2
25	26	Buenos Aires	Argentina	14.5
26	27	Franca	Brazil	13.7
27	28	New York	USA	13.5
28	29	Cape Town	South Africa	13.4
29	30	London	United Kingdom	12.7
30	31	São Paulo	Brazil	11.7
31	32	Quito	Ecuador	10.8
32	33	Kobe	Japan	9.4
33	34	Paris	France	8.8

In [47]:
```
gvi_pivot = pd.pivot_table(
    df,
    values='GVI',
```

```
    index='Country',
    aggfunc=['mean', 'min', 'sum']
)
```

In [56]:
```
gvi_pivot.columns = ['GVI_Average', 'GVI_Minimum', 'GVI_Sum']
print(gvi_pivot)
```

Country	GVI_Average	GVI_Minimum	GVI_Sum
Argentina	14.500000	14.5	14.5
Australia	25.900000	25.9	25.9
Brazil	12.700000	11.7	25.4
Canada	23.633333	19.5	70.9
Ecuador	10.800000	10.8	10.8
France	8.800000	8.8	8.8
Germany	21.500000	21.5	21.5
Israel	17.500000	17.5	17.5
Italy	16.200000	16.2	16.2
Japan	9.400000	9.4	9.4
Mexico	19.800000	16.5	79.2
Netherlands	24.950000	20.6	49.9
Norway	28.800000	28.8	28.8
Singapore	29.300000	29.3	29.3
South Africa	20.233333	13.4	60.7
Switzerland	21.400000	21.4	21.4
USA	21.412500	13.5	171.3
United Kingdom	12.700000	12.7	12.7

Average of GVI of "USA"

In [49]:
```
usa_avg = gvi_pivot.loc['USA', 'GVI_Average']
print("Average GVI of USA:",usa_avg)
```

```
Average GVI of USA: 21.4125
```

Minimum of GVI of "Netherlands"

In [50]:
```
netherlands_min = gvi_pivot.loc['Netherlands', 'GVI_Minimum']
print("Minimum GVI of Netherlands:",netherlands_min)
```

```
Minimum GVI of Netherlands: 20.6
```

Sum of GVI[cities of "Italy"]

In [44]:
```
italy_sum = gvi_pivot.loc['Italy', 'GVI_Sum']

print("Sum of GVI [cities of Italy]:",italy_sum)
```

```
Sum of GVI [cities of Italy]: 16.2
```

The Country with Maximum Cities Top 30

In [51]:
```
# Create a new DataFrame subset containing only rows ranked 1 to 30
top_30_df = df[df['Rank'] <= 30]
```

In [55]:
```
country_counts = top_30_df['Country'].value_counts()

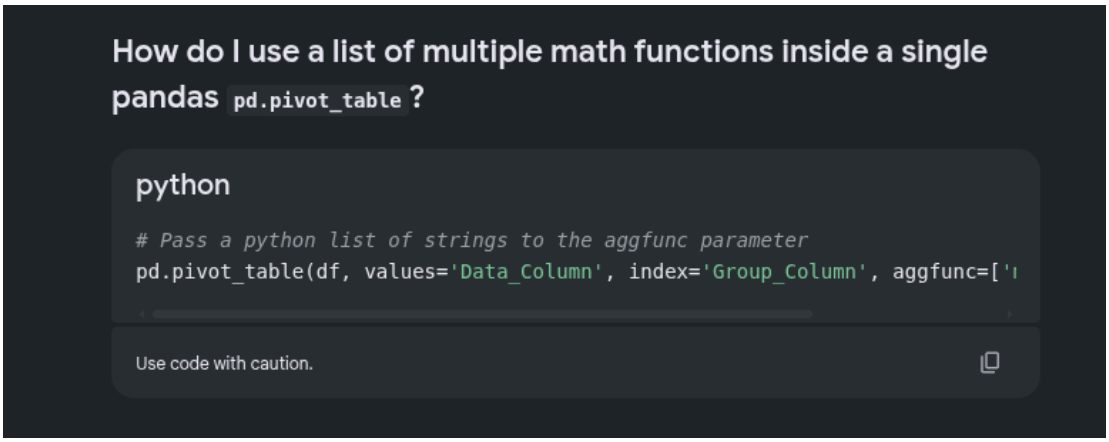
max_country = country_counts.idxmax()
max_cities_count = country_counts.max()

# Print the final result to the console
print("Country with the maximum cities in the Top 30:",max_country)
print("Count of the Country with maximum cities:",max_cities_count)
```

```
Country with the maximum cities in the Top 30: USA
Count of the Country with maximum cities: 8
```

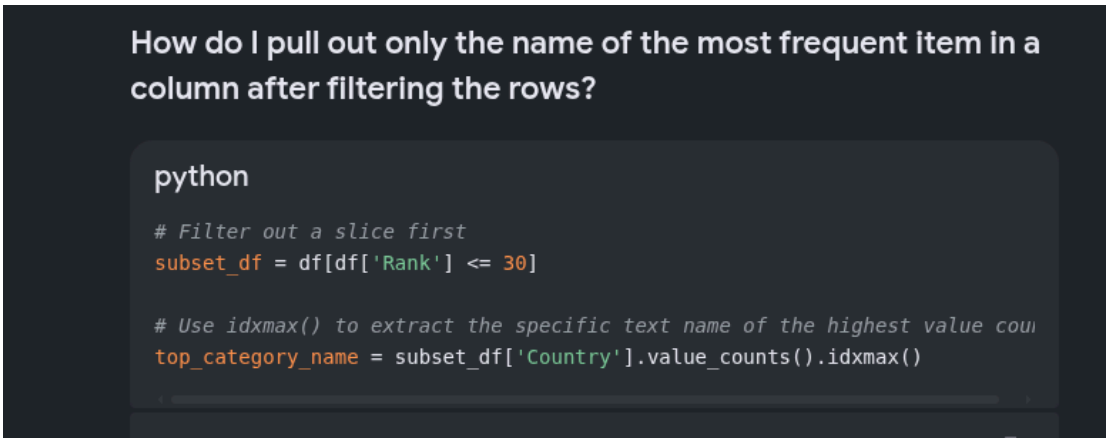
In [ ]:

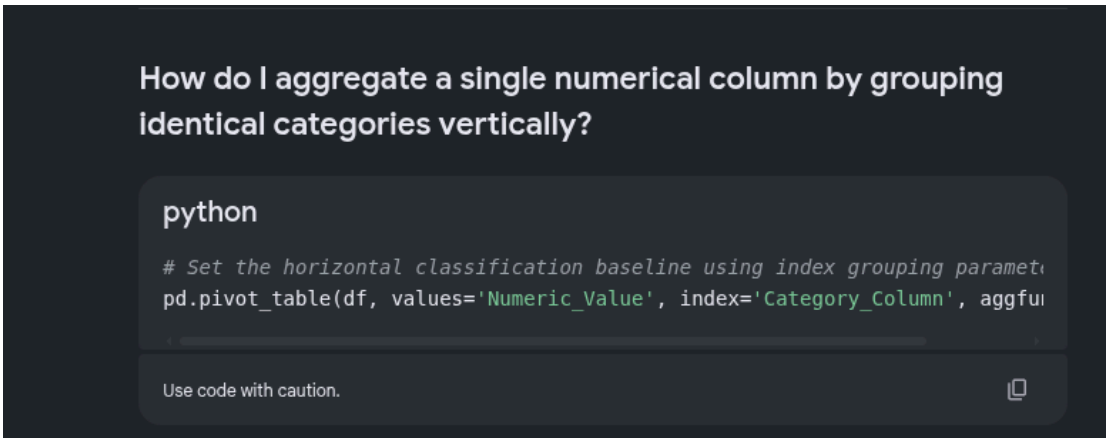
In [ ]:








How do I pull out only the name of the most frequent item in a column after filtering the rows?
python
Filter out a slice first
subset_df = df[df['Rank'] <= 30]
Use idxmax() to extract the specific text name of the highest value cou
top_category_name = subset_df['Country'].value_counts().idxmax()


How do I aggregate a single numerical column by grouping identical categories vertically?
python
Set the horizontal classification baseline using index grouping paramet
pd.pivot_table(df, values='Numeric_Value', index='Category_Column', aggfu
Use code with caution.

In [ ]: